

ATHARVA RAJENDRA DHEMARE

AI/ML Engineer

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SUMMARY

Third-year MBA Tech (Computer Engineering) student at NMIMS specializing in Machine Learning and Computer Vision. Experienced in developing, fine-tuning, and benchmarking deep learning models (YOLOv8, RF-DETR) and building end-to-end video processing pipelines with Python, PyTorch, and OpenCV.

TECHNICAL SKILLS

Languages: Python, C++, Java, SQL

ML / DL Frameworks: PyTorch, scikit-learn, TensorFlow/Keras, OpenCV, Ultralytics YOLOv8, Hugging Face Transformers

Data & Workflow: NumPy, Pandas, Git/GitHub, Google Colab, Kaggle, REST APIs, Linux, n8n

ML Concepts: Computer Vision, Object Detection, Object Tracking, CNNs, Image Processing, Model Training & Evaluation, OCR, NLP/LLMs

EXPERIENCE

Codvo AI — AI/ML Engineer Intern

May '26 – July '26

- Developed and fine-tuned computer-vision models (YOLOv8) for real-time face and object detection, building end-to-end video processing pipelines from data ingestion through inference.
- Built data preprocessing and annotation pipelines—including Pascal VOC / YOLO / COCO format conversion and multi-source dataset merging—to prepare training data for detection models.
- Benchmarked and optimized model inference (FP16, image-size tuning, confidence thresholding) across CPU and GPU, improving real-time throughput and evaluating models on mAP, precision, recall, and FPS.

Girija Welfare Association — Educator (Volunteer Internship)

Apr '25 – May '25

- Conducted interactive learning sessions in Science, English, and Mathematics for underprivileged children, improving conceptual understanding and academic confidence.

PROJECTS

Real-Time Video Analytics System

Python, OpenCV, YOLOv8, ByteTrack

- Built a real-time pipeline that detects and tracks people in video using YOLOv8 and ByteTrack, maintaining persistent track IDs across frames.
- Implemented analytics—virtual line-crossing counting, zone monitoring, and loitering detection—with structured event logging to CSV; debugged identity switches and ghost detections to improve tracking reliability.

Face & Mask Detection — Model Benchmarking

Python, PyTorch, Ultralytics, RF-DETR

- Fine-tuned YOLOv8 on a merged face/mask dataset and engineered the full data pipeline (annotation conversion, train/validation splits, label verification).
- Benchmarked YOLOv8, RetinaFace, and RF-DETR on mAP@0.5, precision, recall, and FPS to identify the best speed–accuracy trade-off for real-time deployment.

KMRL AI Platform

Python, Hugging Face, OCR, n8n

- Implemented OCR with Hugging Face's MinerU model to extract text, tables, and structured data from documents for a Kochi Metro Rail document-management system.
- Designed and deployed an n8n workflow to automate email file extraction and store retrieved documents in the database.

EDUCATION

Narsee Monjee Institute of Management Studies (NMIMS), Navi Mumbai

Jul 2023 – Jul 2028

MBA Tech in Computer Engineering | GPA: 3.4

Delhi World Public School, Nerul — Higher Secondary Education

Apr 2022

Grade: 71%

Bal Bharati Public School, Kharghar — Secondary Education

Apr 2020

Grade: 87%